21 Oct 2026 00:35:38 - psychrolib_psychrometrics.sm
Created using a free version of SMath Studio

ASHRAE 2017 Moist Air Psychrometrics & HVAC Engineering Calculations

2026-10-20

ASHRAE Handbook — Fundamentals (2017) Moist Air Psychrometrics Plugin for SMath Studio

High-Precision Analytical & Iterative Formulations for Building HVAC, Meteorology & Process Design

Native Assembly: PsychroLibWrapper.dll (Plugin GUID: 36a0265c-b529-4440-88e1-88f9226cee6a, v1.2.0.0)

Strict SI Unit Guarantee: Pressure [Pa], Temperature [K], Enthalpy [J/kg] (m^2/s^2), Volume [m³/kg], Density [kg/m³]

The ASHRAE 2017 formulation represents the gold standard in building environmental engineering, meteorological science, and industrial drying/conditioning processes. It models moist air as an ideal mixture of dry air and water vapor, accurately accounting for the temperature dependence of saturation vapor pressure over liquid water (> 0.01 °C) and ice (≤ 0.01 °C), barometric pressure variation with altitude, and non-linear enthalpy-humidity relations.

Key Architectural Highlights of the PsychroLib SMath Plugin (v1.2.0.0 .NET 2.0 CLR):

1. 100% Pure Managed Execution: Zero external C runtimes or native wrappers; runs under Mono 6.14.1 on Linux.
2. Complete SI Standardization: All inputs/outputs strictly anchored to base SI units.
3. Bidirectional Temperature Transparency: Inputs accept Kelvin [K] or Celsius [°C] transparently; outputs strictly return thermodynamic temperature in Kelvin [K].
4. Comprehensive 40-Function API: Clean Psy_<Property>_<Inputs> nomenclature covering dry air, saturated air, moist air mixtures, altimetry, and backward inversions.

Section 1: Standard Atmosphere Altimetry & Barometric Reductions

Evaluation of ambient barometric pressure and temperature profiles from sea level to elevated sites (z = 1200 m).

HVAC project site elevation above mean sea level: 1200 m

$z_site := 1200\ m$

Standard atmospheric pressure at 1200 m: 87715.52 Pa (0.877 bar)

$p_atm_site := \text{Psy_AtmPressure}(z_site) = 87715.5216\ Pa$

Standard atmospheric temperature at 1200 m: 280.35 K (7.20 °C)

$T_atm_site := \text{Psy_AtmTemperature}(z_site) = 280.35\ K$

Measured weather station barometric pressure: 87500 Pa (87.5 kPa)

$p_stn_meas := 87500\ Pa$

Not for commercial use

Created using a free version of SMath Studio

Measured station ambient temperature: 293.15 K (20.0 °C)

$T_stn_meas := 293.15\ K$

Reduced sea-level barometric pressure: 100449.25 Pa (100.45 kPa)

$p_sl_calc := \text{Psy_SeaLevelPressure}(p_stn_meas, z_site, T_stn_meas) = 1.0045 \cdot 10^{5}\ Pa$

Inverse verification of station pressure from sea level: 87500.00 Pa

$p_stn_verify := \text{Psy_StationPressure}(p_sl_calc, z_site, T_stn_meas) = 87500\ Pa$

Section 2: Reference Air State & Psychrometric Properties Evaluation

Complete thermodynamic characterization of summer outdoor air at 35 °C dry-bulb and 60% relative humidity.

Standard sea-level atmospheric pressure: 101325 Pa (1.01325 bar)

$p_ref := 101325\ Pa$

Summer design dry-bulb temperature: 308.15 K (35.0 °C)

$T_db_ref := 308.15\ K$

Design relative humidity: 0.60 (60%)

$RH_ref := 0.60$

Saturation vapor pressure at 35 °C: 5627.82 Pa (5.628 kPa)

$p_ws_ref := \text{Psy_SatVapPres}(T_db_ref) = 5627.8194\ Pa$

Partial water vapor pressure: 3376.69 Pa (3.377 kPa)

$p_v_ref := \text{Psy_VapPres_RH}(T_db_ref, RH_ref) = 3376.6917\ Pa$

Humidity ratio W: 0.021441 kg_H2O/kg_dry_air (21.44 g/kg)

$W_ref := \text{Psy_HumRatio_RH}(T_db_ref, RH_ref, p_ref) = 0.0214$

Specific humidity SH = W/(1+W): 0.020991 kg_H2O/kg_moist_air

$SH_ref := \text{Psy_SpecHum}(W_ref) = 0.021$

Dew-point temperature Tdp: 299.22 K (26.07 °C)

$T_dp_ref := \text{Psy_TDewPoint_RH}(T_db_ref, RH_ref) = 299.218\ K$

Thermodynamic wet-bulb temperature Twb: 301.32 K (28.17 °C)

$T_wb_ref := \text{Psy_TWetBulb_RH}(T_db_ref, RH_ref, p_ref) = 301.3248\ K$

Moist air specific enthalpy h: 90229.93 J/kg (90.23 kJ/kg)

$h_ref := \text{Psy_Enthalpy}(T_db_ref, W_ref) = 90229.9288\ Gy$

Moist air specific volume v: 0.90305 m³/kg_dry_air

$v_ref := \text{Psy_AirVolume}(T_db_ref, W_ref, p_ref) = 0.903\ \frac{m^3}{kg}$

Not for commercial use

Created using a free version of SMath Studio

Moist air density rho: 1.1311 kg/m³

$rho_ref := \text{Psy_AirDensity}(T_db_ref, W_ref, p_ref) = 1.1311 \ \frac{kg}{m^3}$

Degree of saturation mu = W/Wsat: 0.5862

$mu_ref := \text{Psy_DegSaturation}(T_db_ref, W_ref, p_ref) = 0.5862$

Vapor pressure deficit VPD = pws - pv: 2251.13 Pa (2.251 kPa)

$VPD_ref := \text{Psy_VPD}(T_db_ref, W_ref, p_ref) = 2251.1278 \ Pa$

Dry air sensible enthalpy: 35210.00 J/kg (35.21 kJ/kg)

$h_da_ref := \text{Psy_DryAirEnthalpy}(T_db_ref) = 35210 \ Gy$

Dry air specific volume: 0.87295 m³/kg

$v_da_ref := \text{Psy_DryAirVolume}(T_db_ref, p_ref) = 0.873 \ \frac{m^3}{kg}$

Dry air density: 1.1455 kg/m³

$rho_da_ref := \text{Psy_DryAirDensity}(T_db_ref, p_ref) = 1.1455 \ \frac{kg}{m^3}$

Saturation humidity ratio at 35 °C: 0.036576 kg/kg (36.58 g/kg)

$W_sat_ref := \text{Psy_SatHumRatio}(T_db_ref, p_ref) = 0.0366$

Saturation moist air enthalpy: 129066.98 J/kg (129.07 kJ/kg)

$h_sat_ref := \text{Psy_SatEnthalpy}(T_db_ref, p_ref) = 1.2907 \cdot 10^5 \ Gy$

Inverted dry-bulb temperature from enthalpy and humidity ratio: 308.15 K (35.0 °C)

$T_db_inv_h := \text{Psy_TDryBulb_Enthalpy}(h_ref, W_ref) = 308.15 \ K$

Inverted dry-bulb temperature from specific volume: 308.15 K (35.0 °C)

$T_db_inv_v := \text{Psy_TDryBulb_Volume}(v_ref, W_ref, p_ref) = 308.15 \ K$

Section 3: Cross-Property Inversion & State Verification

Verification of inverse relations and cross-dependencies across the 40 PsychroLib functions.

Temperature in Celsius reading: 25 °C

$T_C_test := 25$

Temperature conversion helper: Psy_CelsiusToKelvin(25) = 298.15 K

$T_K_from_C := \text{Psy_CelsiusToKelvin}(T_C_test) = 298.15 \ K$

Inverse temperature conversion helper: Psy_KelvinToCelsius(298.15) = 25.0 °C

$T_C_from_K := \text{Psy_KelvinToCelsius}(T_K_from_C) = 25$

Recover relative humidity from humidity ratio: Psy_RH_HumRatio = 0.60 (60% exact)

$RH_from_W := \text{Psy_RH_HumRatio}(T_db_ref, W_ref, p_ref) = 0.6$

Not for commercial use

Created using a free version of SMath Studio

Recover relative humidity from wet-bulb temperature: Psy_RH_TWetBulb = 0.60

$RH_from_Twb := \text{Psy_RH_TWetBulb}(T_db_ref, T_wb_ref, p_ref) = 0.6$

Recover relative humidity from dew-point temperature: Psy_RH_TDewPoint = 0.60

$RH_from_Tdp := \text{Psy_RH_TDewPoint}(T_db_ref, T_dp_ref) = 0.6$

Recover relative humidity from vapor pressure: Psy_RH_VapPres = 0.60

$RH_from_pv := \text{Psy_RH_VapPres}(T_db_ref, p_v_ref) = 0.6$

Recover humidity ratio from wet-bulb temperature: 0.021441 kg/kg

$W_from_Twb := \text{Psy_HumRatio_TWetBulb}(T_db_ref, T_wb_ref, p_ref) = 0.0214$

Recover humidity ratio from dew-point temperature: 0.021441 kg/kg

$W_from_Tdp := \text{Psy_HumRatio_TDewPoint}(T_dp_ref, p_ref) = 0.0214$

Recover humidity ratio from enthalpy: 0.021441 kg/kg

$W_from_h := \text{Psy_HumRatio_Enthalpy}(h_ref, T_db_ref) = 0.0214$

Section 4: HVAC Cooling and Dehumidification Coil Simulation

Air handling unit (AHU) cooling coil simulation conditioning 2.5 kg/s of fresh outdoor air.

Air enters at 32 °C, 65% RH and is cooled below its dew point to an apparatus dew point (ADP) of 10 °C with BF = 0.12.

Dry air mass flow rate: 2.5 kg/s

$m_dot_da := \frac{2.5\ \text{kg}}{\text{s}}$

Entering air dry-bulb temperature: 305.15 K (32.0 °C)

$T_in := 305.15\ \text{K}$

Entering air relative humidity: 0.65 (65%)

$RH_in := 0.65$

Entering humidity ratio W_in: 0.019583 kg/kg (19.58 g/kg)

$W_in := \text{Psy_HumRatio_RH}(T_in, RH_in, p_ref) = 0.0196$

Entering specific enthalpy h_in: 82335.42 J/kg (82.34 kJ/kg)

$h_in := \text{Psy_Enthalpy}(T_in, W_in) = 82335.4176\ \text{Gy}$

Entering dew point: 297.74 K (24.59 °C)

$T_dp_in := \text{Psy_TDewPoint_RH}(T_in, RH_in) = 297.7425\ \text{K}$

Coil Apparatus Dew Point (ADP) surface temperature: 283.15 K (10.0 °C)

$T_adp := 283.15\ \text{K}$

Saturation humidity ratio at coil ADP: 0.007630 kg/kg (7.63 g/kg)

$W_adp := \text{Psy_SatHumRatio}(T_adp, p_ref) = 0.0076$

Not for commercial use

Created using a free version of SMath Studio

Saturation enthalpy at coil ADP: 29284.68 J/kg (29.28 kJ/kg)

$h_adp := \text{Psy_SatEnthalpy}(T_adp, p_ref) = 29284.6833\ \text{Gy}$

Coil bypass factor BF: 0.12 (Contact Factor CF = 0.88)

$BF := 0.12$

Leaving air dry-bulb temperature T_out: 285.79 K (12.64 °C)

$T_out := T_adp + BF \cdot (T_in - T_adp) = 285.79\ \text{K}$

Leaving air humidity ratio W_out: 0.009064 kg/kg (9.06 g/kg)

$W_out := W_adp + BF \cdot (W_in - W_adp) = 0.0091$

Leaving air specific enthalpy h_out: 35599.12 J/kg (35.60 kJ/kg)

$h_out := \text{Psy_Enthalpy}(T_out, W_out) = 35599.1197\ \text{Gy}$

Leaving air relative humidity RH_out: 0.9950 (99.50% near-saturated)

$RH_out := \text{Psy_RH_HumRatio}(T_out, W_out, p_ref) = 0.995$

Leaving air dew point Tdp_out: 285.71 K (12.56 °C)

$T_dp_out := \text{Psy_TDewPoint_HumRatio}(T_out, W_out, p_ref) = 285.7131\ \text{K}$

Moisture condensation removal rate: 0.026297 kg/s

$m_dot_w := m_dot_da \cdot (W_in - W_out) = 0.0263\ \frac{\text{kg}}{\text{s}}$

Condensate drainage volume rate: 94.67 L/hr

$$V_dot_condense := \frac{\frac{m_dot_w}{\frac{1000\ \text{kg}}{\text{m}^3}} \cdot 3600\ \text{s}}{\frac{\text{m}^3}{\text{hr}}} \cdot 1000\ \text{L}$$

Total coil cooling capacity: 116840.74 W (116.84 kW)

$q_total := m_dot_da \cdot (h_in - h_out) = 1.1684 \cdot 10^5\ \text{W}$

Sensible cooling load: 50453.37 W (50.45 kW)

$$q_sens := m_dot_da \cdot \left(\frac{1006\ \text{J}}{\frac{\text{kg}}{\text{K}}} + \frac{1860\ \text{J}}{\frac{\text{kg}}{\text{K}}} \cdot W_in \right) \cdot (T_in - T_out) = 50453.3665\ \text{W}$$

Latent dehumidification load: 66387.38 W (66.39 kW)

$q_latent := q_total - q_sens = 66387.3784\ \text{W}$

Sensible Heat Ratio SHR = q_sens / q_total: 0.4318 (43.18% sensible)

$$SHR := \frac{q_sens}{q_total} = 0.4318$$

Not for commercial use

Created using a free version of SMath Studio

Section 5: Direct Evaporative Cooling in Arid Climate

Simulation of direct evaporative cooling (swamp cooler) operating in hot, desert ambient conditions.

Intake air at 40 °C, 15% RH is adiabatically humidified across a media pack with 85% saturation effectiveness.

Desert ambient dry-bulb temperature: 313.15 K (40.0 °C)

$T_dry_amb := 313.15\ K$

Desert ambient relative humidity: 0.15 (15%)

$RH_dry_amb := 0.15$

Ambient humidity ratio W_dry: 0.006873 kg/kg (6.87 g/kg)

$W_dry_amb := \text{Psy_HumRatio_RH}(T_dry_amb, RH_dry_amb, p_ref) = 0.0069$

Ambient air enthalpy h_dry: 57941.27 J/kg (57.94 kJ/kg)

$h_dry_amb := \text{Psy_Enthalpy}(T_dry_amb, W_dry_amb) = 57941.2679\ Gy$

Ambient wet-bulb limit Twb: 293.51 K (20.36 °C)

$T_wb_dry := \text{Psy_TWetBulb_RH}(T_dry_amb, RH_dry_amb, p_ref) = 293.5056\ K$

Evaporative media saturation effectiveness: 0.85 (85%)

$eta_evap := 0.85$

Supply air dry-bulb temperature: 296.45 K (23.30 °C)

$T_evap_out := T_dry_amb - eta_evap \cdot (T_dry_amb - T_wb_dry) = 296.4523\ K$

Supply air humidity ratio (constant enthalpy process): 0.013559 kg/kg (13.56 g/kg)

$W_evap_out := \text{Psy_HumRatio_Enthalpy}(h_dry_amb, T_evap_out) = 0.0136$

Supply air relative humidity: 0.7553 (75.53% comfortable)

$RH_evap_out := \text{Psy_RH_HumRatio}(T_evap_out, W_evap_out, p_ref) = 0.7553$

Supply air dew point: 291.89 K (18.74 °C)

$T_dp_evap_out := \text{Psy_TDewPoint_HumRatio}(T_evap_out, W_evap_out, p_ref) = 291.8863\ K$

Temperature reduction achieved by evaporative cooling: 16.70 K (16.70 °C cooling)

$delta_T_cool := T_dry_amb - T_evap_out = 16.6977\ K$

Section 6: High-Resolution 2D Visualizations & Psychrometric Chart Curves

Interactive engineering charts rendered via Chart2DLib (XYPlotRegion): saturation vapor pressure, humidity ratio psychrometric chart curves, moist air enthalpy, and atmospheric altimetry profile.

Not for commercial use

Created using a free version of SMath Studio

Visualization 1: Saturation Vapor Pressure Curve pws(T)

ASHRAE 2017 equations 5 (ice regime <= 0.01 °C) and 6 (water regime > 0.01 °C) spanning -20 °C to +50 °C.

Temperature nodes: [-20 ... +50] °C (253.15 ... 323.15 K)

$$T_pws_nodes := \begin{bmatrix} 253.15 \\ 258.15 \\ 263.15 \\ 268.15 \\ 273.15 \\ 278.15 \\ 283.15 \\ 288.15 \\ 293.15 \\ 298.15 \\ 303.15 \\ 308.15 \\ 313.15 \\ 318.15 \\ 323.15 \end{bmatrix} \text{K}$$

Saturation vapor pressure values: [103.28 ... 12351.4] Pa

$$p_ws_curve := \overrightarrow{\text{Psy_SatVapPres}(T_pws_nodes)}$$

Saturation vapor pressure in kPa: [0.103 ... 12.35] kPa

$$p_ws_kPa := \frac{\overrightarrow{p_ws_curve}}{1000}$$

Temperature in Celsius: [-20 ... +50] °C

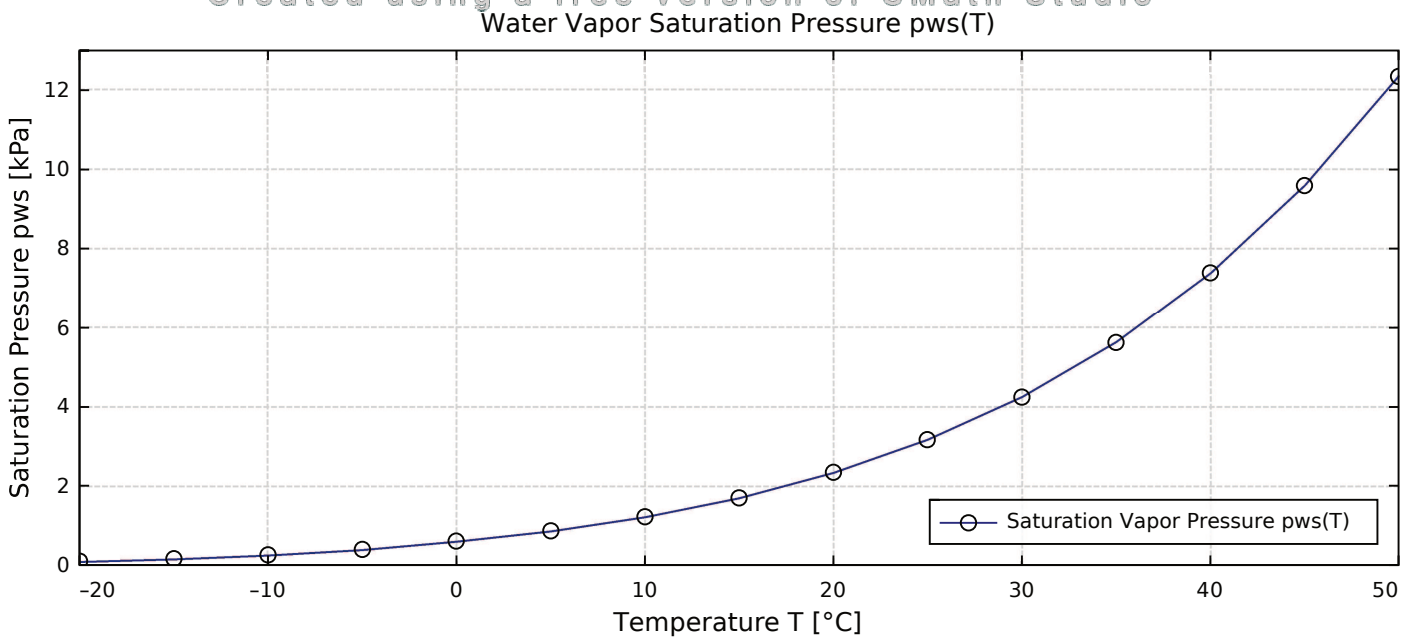
$$T_pws_C := \overrightarrow{T_pws_nodes - 273.15\ \text{K}}$$

XY coordinates for saturation vapor pressure plot

$$xy_pws_curve := \text{augment}(T_pws_C, p_ws_kPa)$$

Not for commercial use

Created using a free version of SMath Studio



{xy_pws_curve

Visualization 2: Psychrometric Chart Skeleton — Humidity Ratio W(T) vs Temperature

Constant relative humidity isolines at standard sea-level atmospheric pressure (p = 101325 Pa).

Chart temperature nodes: [0 ... 45] °C (273.15 ... 318.15 K)

$$T_chart_nodes := \begin{bmatrix} 273.15 \\ 278.15 \\ 283.15 \\ 288.15 \\ 293.15 \\ 298.15 \\ 303.15 \\ 308.15 \\ 313.15 \\ 318.15 \end{bmatrix} K$$

Chart temperatures in Celsius: [0 ... 45] °C

$$T_chart_C := \overrightarrow{T_chart_nodes - 273.15\ K}$$

100% RH saturation line humidity ratio: [3.77 ... 65.04] g/kg

$$w_100_curve := \overrightarrow{Psy_SatHumRatio(T_chart_nodes, 101325\ Pa)} \cdot 1000$$

75% RH isoline humidity ratio: [2.83 ... 47.54] g/kg

$$w_75_curve := \overrightarrow{Psy_HumRatio_RH(T_chart_nodes, 0.75, 101325\ Pa)} \cdot 1000$$

50% RH isoline humidity ratio: [1.88 ... 30.91] g/kg

$$w_50_curve := \overrightarrow{Psy_HumRatio_RH(T_chart_nodes, 0.50, 101325\ Pa)} \cdot 1000$$

Not for commercial use

Created using a free version of SMath Studio

25% RH isoline humidity ratio: [0.94 ... 15.08] g/kg

$w_25_curve := \overrightarrow{\text{Psy_HumRatio_RH}(T_chart_nodes, 0.25, 101325\ \text{Pa})} \cdot 1000$

100% RH saturation curve coordinates

$xy_w_100 := \text{augment}(T_chart_C, w_100_curve)$

75% RH isoline coordinates

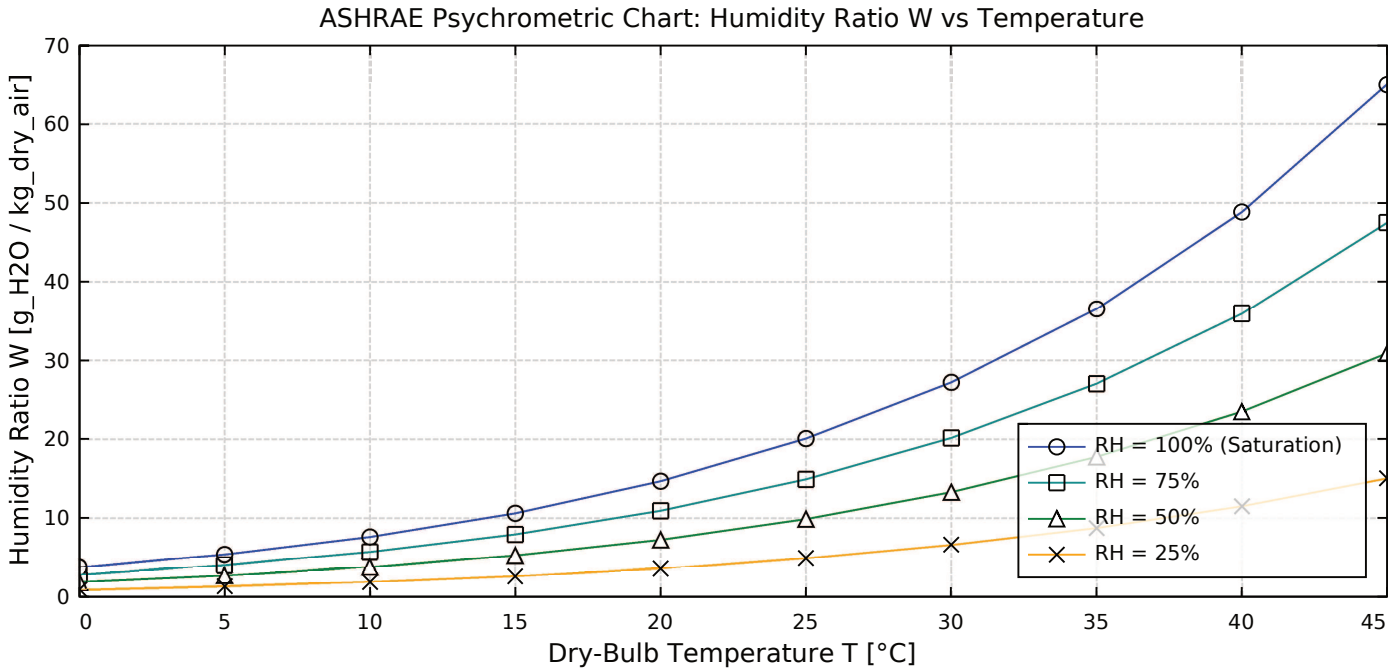
$xy_w_75 := \text{augment}(T_chart_C, w_75_curve)$

50% RH isoline coordinates

$xy_w_50 := \text{augment}(T_chart_C, w_50_curve)$

25% RH isoline coordinates

$xy_w_25 := \text{augment}(T_chart_C, w_25_curve)$



$$\begin{cases} xy_w_100 \\ xy_w_75 \\ xy_w_50 \\ xy_w_25 \end{cases}$$

Visualization 3: Moist Air Enthalpy Isotherms h(T)

Specific enthalpy profiles comparing saturated moist air (100% RH), typical indoor/outdoor air (50% RH), and dry air (0% RH).

100% RH saturated air enthalpy: [9.44 ... 213.39] kJ/kg

$h_100_kJ := \frac{\overrightarrow{\text{Psy_SatEnthalpy}(T_chart_nodes, 101325\ \text{Pa})}}{1000}$

Not for commercial use

Created using a free version of SMath Studio

50% RH moist air enthalpy: [4.71 ... 125.15] kJ/kg

$$h_50_kJ := \frac{\overrightarrow{\text{Psy_Enthalpy}\left(T_chart_nodes, \frac{w_50_curve}{1000}\right)}}{1000}$$

Dry air sensible enthalpy: [0.00 ... 45.27] kJ/kg

$$h_0_kJ := \frac{\overrightarrow{\text{Psy_DryAirEnthalpy}\left(T_chart_nodes\right)}}{1000}$$

100% RH enthalpy coordinates

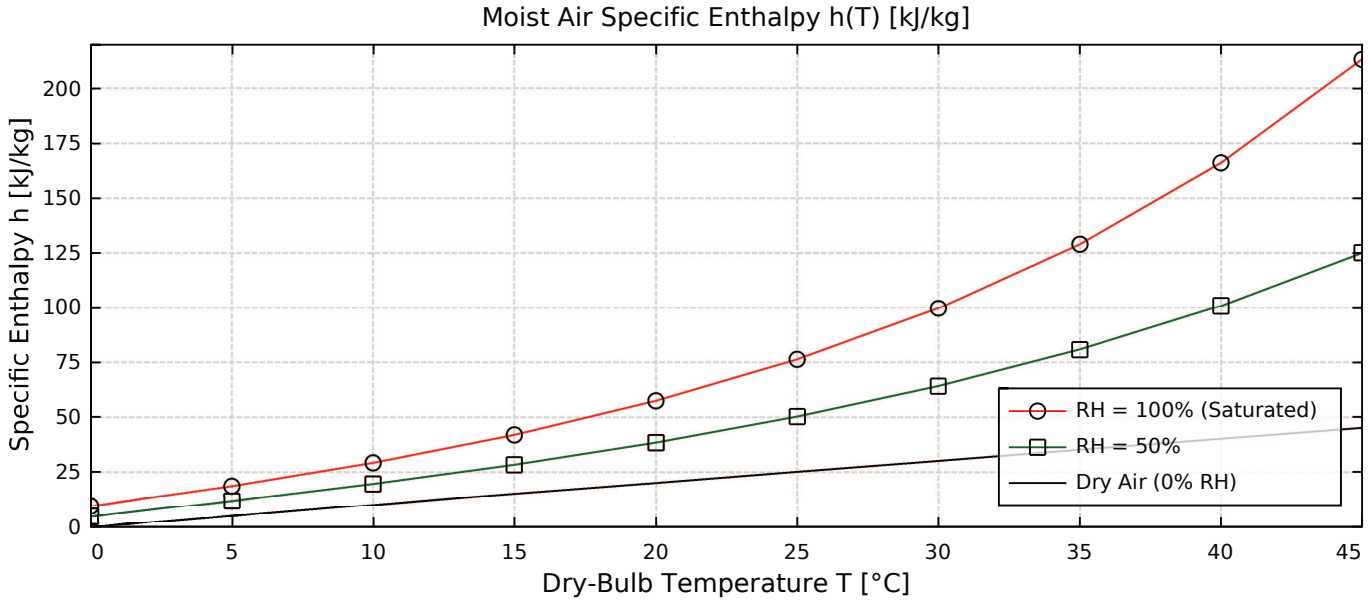
$$xy_h_100 := \text{augment}\left(T_chart_C, h_100_kJ\right)$$

50% RH enthalpy coordinates

$$xy_h_50 := \text{augment}\left(T_chart_C, h_50_kJ\right)$$

0% RH dry air enthalpy coordinates

$$xy_h_0 := \text{augment}\left(T_chart_C, h_0_kJ\right)$$



$$\begin{cases} xy_h_100 \\ xy_h_50 \\ xy_h_0 \end{cases}$$

Visualization 4: Standard Atmosphere Barometric Profile p(z)

ASHRAE 2017 equation 3 standard atmospheric barometric pressure reduction from sea level up to 5000 m.

Not for commercial use

Created using a free version of SMath Studio

Altimetry nodes: [0 ... 5000] m

$$z_nodes := \begin{bmatrix} 0 \\ 500 \\ 1000 \\ 1500 \\ 2000 \\ 2500 \\ 3000 \\ 3500 \\ 4000 \\ 4500 \\ 5000 \end{bmatrix} \text{ m}$$

Atmospheric pressure values: [101325 ... 54020] Pa

$$p_alt_curve := \overrightarrow{\text{Psy_AtmPressure}(z_nodes)}$$

Atmospheric pressure in kPa: [101.33 ... 54.02] kPa

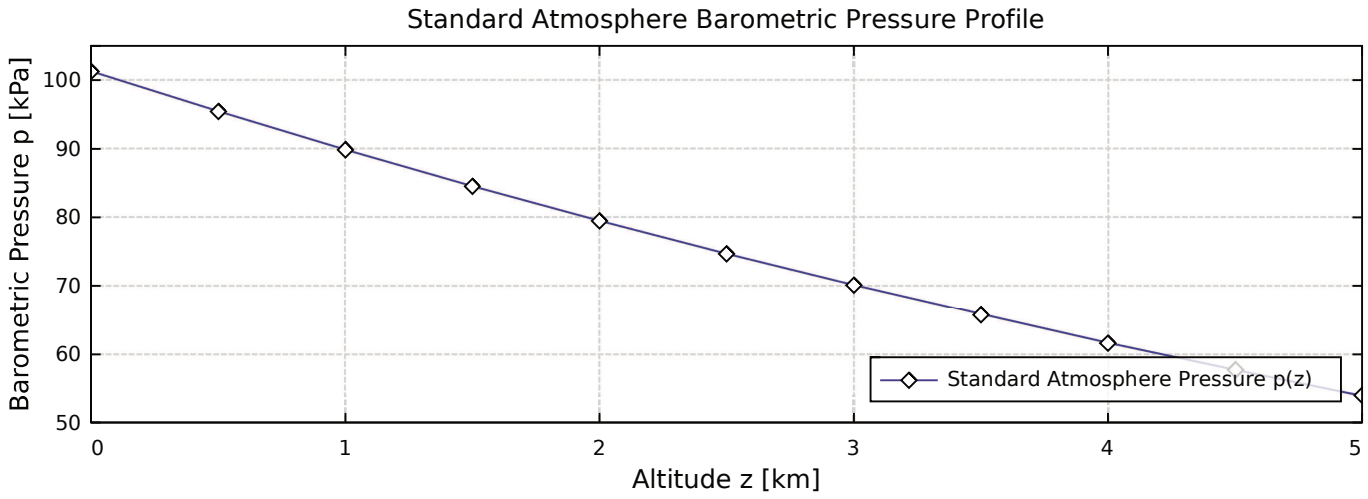
$$p_alt_kPa := \frac{\overrightarrow{p_alt_curve}}{1000}$$

Elevation in km: [0 ... 5] km

$$z_km := \frac{\overrightarrow{z_nodes}}{1000}$$

Altimetry pressure profile coordinates

$$xy_alt_curve := \text{augment}(z_km, p_alt_kPa)$$



{xy_alt_curve

Not for commercial use